Chapter 5 Plane Waves

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5.1 Electromagnetic Plane Waves

In this section, we consider the frequency domain source-free Maxwell equations in lossless medium

$$\vec{\nabla} \times \vec{E} = -i\omega\mu\vec{H} \tag{5.1.1a}$$

$$\vec{\nabla} \times \vec{H} = i\omega\epsilon\vec{E} \tag{5.1.1b}$$

$$\vec{\nabla} \cdot \vec{E} = 0 \tag{5.1.1c}$$

$$\vec{\nabla} \cdot \vec{H} = 0 \tag{5.1.1d}$$

The corresponding wave equation for the electric field

$$\nabla^2\vec{E} + k^2\vec{E} = 0 \tag{5.1.2}$$

with $k = \omega\sqrt{\mu\epsilon}$ (wavenumber).

5.1.1 Plane Wave Solution

To acquire the plane wave solution of (5.1.2), we consider the Cartesian coordinates for analysis

$$\nabla^2\vec{E} + k^2\vec{E} = 0 =$$

$$= \nabla^2\left(E_x\hat{x} + E_y\hat{y} + E_z\hat{z}\right) + k^2\left(E_x\hat{x} + E_y\hat{y} + E_z\hat{z}\right) \tag{5.1.3}$$

$$= \hat{x}(\nabla^2 E_x + k^2 E_x) + \hat{y}\left(\nabla^2 E_y + k^2 E_y\right) + \hat{z}(\nabla^2 E_z + k^2 E_z)$$

implying each component of the electric field satisfy the Helmholtz equation

$$\nabla^2\psi + k^2\psi = 0 \tag{5.1.4}$$

With $\nabla^2 = \partial_x^2 + \partial_y^2 + \partial_z^2$, and by separation of variables, we can write

$$\psi(x, y, z) = X(x)Y(y)Z(z)$$

Plugging back to (5.1.4), we get

$$\frac{1}{X}\frac{d^2X}{dx^2} + \frac{1}{Y}\frac{d^2Y}{dy^2} + \frac{1}{Z}\frac{d^2Z}{dz^2} + k^2 = 0 \tag{5.1.5}$$

Thus, each term in (5.1.5) should be a constant value. We can write

$$\begin{cases} \dfrac{d^2X}{dx^2} + k_x^2 X = 0 \\ \dfrac{d^2Y}{dy^2} + k_y^2 Y = 0 \\ \dfrac{d^2Z}{dz^2} + k_z^2 Z = 0 \end{cases} \tag{5.1.6}$$

With the relation

$$k_z^2 + k_y^2 + k_x^2 = k^2 \tag{5.1.7}$$

The elementary solution for $\psi(x, y, z)$ is

$$\psi = e^{\pm i(k_x x + k_y y + k_z z)} = e^{\pm i\vec{k}\cdot\vec{r}} \tag{5.1.8}$$

where $\vec{k} = k_x\hat{x} + k_y\hat{y} + k_z\hat{z}$ is the wave vector, denoting the propagation direction of the plane wave. Transforming the phasor into time domain (with $e^{i\omega t}$convention), we can show that $e^{i\vec{k}\cdot\vec{r}}$ is plane wave propagating in $-\vec{k}$ direction and $e^{-i\vec{k}\cdot\vec{r}}$ is plane wave propagating in $\vec{k}$ direction.

We consider $e^{-i\vec{k}\cdot\vec{r}}$ for the following discussion. We can write the electric vector field as

$$\vec{E} = \vec{E}_0\, e^{-i\vec{k}\cdot\vec{r}} \tag{5.1.9}$$

where $\vec{E}_0$ is a constant vector. We put (5.1.9) back to (5.1.1c) and use the identity $\vec{\nabla}e^{-i\vec{k}\cdot\vec{r}} = -i\vec{k}e^{-i\vec{k}\cdot\vec{r}}$:

$$\vec{\nabla}\cdot\vec{E} = \vec{\nabla}\cdot\left(\vec{E}_0 e^{-i\vec{k}\cdot\vec{r}}\right) = -ie^{-i\vec{k}\cdot\vec{r}}\left(\vec{k}\cdot\vec{E}_0\right) = 0 \tag{5.1.10}$$

And from (5.1.1a) we get

$$\vec{H} = \frac{i}{\omega\mu}\vec{\nabla}\times\vec{E} = \frac{1}{\omega\mu}\vec{k}\times\vec{E} = \frac{1}{\eta}\hat{k}\times\vec{E} = \vec{H}_0 e^{-i\vec{k}\cdot\vec{r}} \tag{5.1.11}$$

with $\vec{H}_0 = \frac{1}{\eta}\hat{k}\times\vec{E}_0$. (5.1.10) and (5.1.11) imply that

i. $\vec{k}$, $\vec{E}_0$ and $\vec{H}_0$ are penedicular to each other. We call this transverse electromagnetic (TEM) wave.
ii. The ratio of the magnitude of electric field to the magnitude of the magnetic field is $\eta = \sqrt{\mu/\epsilon}$. In free space, $\eta \cong 377$ (Ohm).

Let us consider a plane wave propagte in the +z direction ($\vec{k} = k\hat{z}$), then (5.1.8) reduces to $\psi = e^{-ikz}$. Since $\vec{E}_0$ is perpendicular to the z-axis, we choose $\vec{E}_0 = E_x\hat{x}$ for simplicity. Then from (5.1.11) we have $\vec{H}_0 = \frac{E_x}{\eta}\hat{y}$. Transform the wave into time domain yields

$$\vec{E}(x,y,z,t) = \Re\left(\vec{E}e^{i\omega t}\right) = E_x\cos(\omega t - kz)\hat{x} \tag{5.1.12}$$

This is a traveling wave solution. To find the velocity of the wavefront v_p (phase velocity), we set $\omega t - kz = constant$ and take derivative respect to t

$$\frac{dz}{dt} = v_p = \frac{\omega}{k} = \frac{1}{\sqrt{\mu\epsilon}} \tag{5.1.13}$$

and the wavenumber k has the relationship

$$k = \frac{\omega}{v_p} = \frac{2\pi}{\lambda} \tag{5.1.14}$$

where λ is the wavelength.

□ **<u>Example 5-1</u>** <u>(Cheng 8-1)</u>

A uniform plane wave with an electric field $\vec{E} = E_x\hat{x}$ propagates in a lossless medium characterized by $\epsilon_r = 4$, $\mu_r = 1$, and $\sigma = 0$ in the $+z$-direction. The electric field component E_x is sinusoidal with a frequency of 100 MHz and attains a maximum value of 10^{-4} V/m at $t = 0$ and $z = \frac{1}{8}$ m.

(a) Derive the instantaneous expression for $\vec{E}$ at any t and z.

(b) Determine the corresponding instantaneous expression for $\vec{H}$.

(c) Identify the locations where E_x reaches a positive maximum at $t = 10^{-8}$ s.

Solution

$$k = \omega\sqrt{\mu\epsilon} = \frac{\omega}{c}\sqrt{\mu_r\epsilon_r} = \frac{4\pi}{3}$$

(a)

$$E_x = E_0\cos(\omega t - kz + \psi) = 10^{-4}\cos\left(2\pi \times 10^8 t - \frac{4\pi}{3}z + \psi\right)$$

Maxima occurs at $t = 0$ and $z = \frac{1}{8} \Longrightarrow \psi = \frac{4\pi}{3}\frac{1}{8} = \frac{\pi}{6}$

$$\therefore E_x = 10^{-4}\cos\left(2\pi \times 10^8 t - \frac{4\pi}{3}z + \frac{\pi}{6}\right)$$

(b)

$$\eta = \sqrt{\frac{\mu}{\epsilon}} = \frac{\eta_0}{\sqrt{\epsilon_r}} = 60\pi$$

$$\vec{H} = \frac{E_x}{\eta}\hat{y} = \frac{10^{-4}}{60\pi}\cos\left(2\pi \times 10^8 t - \frac{4\pi}{3}z + \frac{\pi}{6}\right)\hat{y}$$

(c)
We get maximum E_x when

$$2\pi \times 10^8 \times 10^{-8} - \frac{4\pi}{3}z + \frac{\pi}{6} = 2n\pi, \qquad n \in \mathbb{Z}$$

$$\therefore z = \frac{13}{8} - \frac{3}{2}n$$

5.1.2 Propagation in Lossy Media

For plane wave propagation in lossy media, the wavenumber k in (5.1.2) is replace by $k_c = \omega\sqrt{\mu\epsilon_c}$, where $\epsilon_c = \epsilon' - i\epsilon''$ is a complex number as in (4.5.4)

$$\nabla^2\vec{E} + k_c^2\vec{E} = 0 \tag{5.1.15}$$

We can define the propagation constant γ as

$$\gamma = ik_c = i\omega\sqrt{\mu\epsilon_c} = \alpha + i\beta, \qquad \alpha, \beta \in \mathbb{R} \tag{5.1.16}$$

where α is called the attenuation constant (Np/m) and β is called the phase constant (rad/m). For a +z propagating plane wave with only x-component electric field

$$E_x = E_0^+ e^{-ik_c z} = E_0^+ e^{-\alpha z} e^{-i\beta z} \tag{5.1.17}$$

where $e^{-\alpha z}$ accounts for the attenuation and $e^{-i\beta z}$ accounts for the propagation. From $\epsilon_c = \epsilon' - \epsilon''$ and $k_c = \beta - i\alpha$, we can get

$$\beta = \omega\left\{\frac{\mu\epsilon'}{2}\left[\sqrt{1+\left(\frac{\epsilon''}{\epsilon'}\right)^2}+1\right]\right\}^{1/2} \tag{5.1.18}$$

$$\alpha = \omega\left\{\frac{\mu\epsilon'}{2}\left[\sqrt{1+\left(\frac{\epsilon''}{\epsilon'}\right)^2}-1\right]\right\}^{1/2} \tag{5.1.19}$$

The derivation can be done by squaring k_c and equate the real and imaginary part. The corresponding complex intrinsic impedance and phase velocity are

$$\eta_c = \sqrt{\frac{\mu}{\epsilon_c}} = \sqrt{\frac{\mu}{\epsilon'}}\left(1 - i\frac{\epsilon''}{\epsilon'}\right)^{-1/2} \tag{5.1.20}$$

$$v_p = \frac{\omega}{\beta} = \left\{\frac{\mu\epsilon'}{2}\left[\sqrt{1+\left(\frac{\epsilon''}{\epsilon'}\right)^2}-1\right]\right\}^{-1/2} \tag{5.1.21}$$

Two limiting cases are considered:

A. Low-loss dielectrics

With low-loss dielectrics we have $\epsilon'' \ll \epsilon'$ (or $\sigma/\epsilon\omega \ll 1$). Under such assumptions we have

$$\begin{aligned}\beta \cong\ \omega\left\{\frac{\mu\epsilon'}{2}\left[1+\frac{1}{2}\left(\frac{\epsilon''}{\epsilon'}\right)^2+1\right]\right\}^{\frac{1}{2}} &= \omega\left\{\mu\epsilon'\left[1+\frac{1}{4}\left(\frac{\epsilon''}{\epsilon'}\right)^2\right]\right\}^{\frac{1}{2}} \\ &= \omega\sqrt{\mu\epsilon'}\left[1+\frac{1}{8}\left(\frac{\epsilon''}{\epsilon'}\right)^2\right]\end{aligned} \tag{5.1.22}$$

$$\alpha \cong\ \omega\left\{\frac{\mu\epsilon'}{2}\left[1+\frac{1}{2}\left(\frac{\epsilon''}{\epsilon'}\right)^2-1\right]\right\}^{\frac{1}{2}} = \frac{\omega\epsilon''}{2}\sqrt{\frac{\mu}{\epsilon'}} \tag{5.1.23}$$

$$\eta_c \cong \sqrt{\frac{\mu}{\epsilon'}}\left(1 + i\frac{\epsilon''}{2\epsilon'}\right) \tag{5.1.24}$$

$$v_p \cong \frac{1}{\sqrt{\mu\epsilon'}}\left[1-\frac{1}{8}\left(\frac{\epsilon''}{\epsilon'}\right)^2\right] \tag{5.1.25}$$

B. Good conductors

With good conductors we have $\epsilon'' \gg \epsilon'$ (or $\sigma/\epsilon\omega \gg 1$). Since we mostly use σ to denote the loss in conductors, we use $\epsilon' = \epsilon$ and $\epsilon'' = \sigma/\omega$ in our derivation. Under such assumptions we have

$$\begin{aligned}\beta \cong \alpha \cong\ \omega\left\{\frac{\mu\epsilon'}{2}\left[\left(\frac{\epsilon''}{\epsilon'}\right)\sqrt{1+\left(\frac{\epsilon'}{\epsilon''}\right)^2}\right]\right\}^{\frac{1}{2}} \\ = \omega\sqrt{\frac{\mu\epsilon''}{2}} = \sqrt{\pi\mu f\sigma}\end{aligned} \tag{5.1.26}$$

$$\eta_c \cong (1 + i)\sqrt{\frac{\pi\mu f}{\sigma}} \tag{5.1.27}$$

$$v_p \cong \sqrt{\frac{2\omega}{\mu\sigma}} \tag{5.1.28}$$

From (5.1.27), we observe that in good conductors, the magnetic field lags approximately 45 degrees behind the electric field. The skin depth is defined as the reciprocal of the α

$$\delta = \frac{1}{\alpha} = \frac{1}{\sqrt{\pi\mu f\sigma}} \tag{5.1.29}$$

whcih is also the distance through which the amplitude of a traveling plane wave decreases by a factor of $e^{-1} \cong 0.368$.

□ **Example 5-2** (Cheng 8-4)

A uniform plane wave with an electric field $\vec{E} = E_x\hat{x}$ propagates in a lossless medium characterized by $\epsilon_r = 4$, $\mu_r = 1$, and $\sigma = 0$ in the $+z$-direction. The electric field component E_x is sinusoidal with a frequency of 100 MHz and attains a maximum value of 10^{-4} V/m at $t = 0$ and $z = \frac{1}{8}$ m.

(a) Derive the instantaneous expression for $\vec{E}$ at any t and z.

(b) Determine the corresponding instantaneous expression for $\vec{H}$.

(c) Identify the locations where E_x reaches a positive maximum at $t = 10^{-8}$ s.

Solution

$$k = \omega\sqrt{\mu\epsilon} = \frac{\omega}{c}\sqrt{\mu_r\epsilon_r} = \frac{4\pi}{3}$$

(a)

$$E_x = E_0\cos(\omega t - kz + \psi) = 10^{-4}\cos\left(2\pi \times 10^8 t - \frac{4\pi}{3}z + \psi\right)$$

Maxima occurs at $t = 0$ and $z = \frac{1}{8} \Longrightarrow \psi = \frac{4\pi}{3}\frac{1}{8} = \frac{\pi}{6}$

$$\therefore E_x = 10^{-4}\cos\left(2\pi \times 10^8 t - \frac{4\pi}{3}z + \frac{\pi}{6}\right)$$

(b)

$$\eta = \sqrt{\frac{\mu}{\epsilon}} = \frac{\eta_0}{\sqrt{\epsilon_r}} = 60\pi$$

$$\vec{H} = \frac{E_x}{\eta}\hat{y} = \frac{10^{-4}}{60\pi}\cos\left(2\pi \times 10^8 t - \frac{4\pi}{3}z + \frac{\pi}{6}\right)\hat{y}$$

(c)

We get maximum E_x when

$$2\pi \times 10^8 \times 10^{-8} - \frac{4\pi}{3}z + \frac{\pi}{6} = 2n\pi, \qquad n \in \mathbb{Z}$$

$$\therefore z = \frac{13}{8} - \frac{3}{2}n$$

5.2 Polarization

Consider a +z-propagated plane wave

$$\vec{E} = \left(E_x\hat{x} + E_y\hat{y}\right)e^{-ikz} \tag{5.2.1}$$

In general, E_x and E_y are complex with the following representation

$$\begin{cases} E_x = |E_x|e^{i\delta_x} \\ E_y = |E_y|e^{i\delta_y} \end{cases} \tag{5.2.2}$$

By the transformation (4.5.1) from phasor to time domain we can characterize the wave with three polarization states:

A. Linear Polarization

For linearly polarized field we have

$$\delta_x - \delta_y = n\pi \tag{5.2.3}$$

with $n \in \mathbb{Z}$. For the time-domain fields, we can check that both the x and y components of the electric field are with same phase.

B. Circular Polarization

For circularly polarized field we have

$$\begin{cases} \delta_x - \delta_y = \dfrac{n\pi}{2} \\ |E_x| = |E_y| \end{cases} \tag{5.2.4}$$

We considered the following representation for simplicity with $E_0 \in \mathbb{R}$ and $e^{i\frac{\pi}{2}} = i$. We can show that one represents a right-hand circularly polarized (RHCP) field, while the other corresponds to a left-hand circularly polarized (LHCP) field (using your thumb as the direction of propagation).

$$\vec{E}_{RC} = (E_0\hat{x} - iE_0\hat{y})e^{-ikz} \tag{5.2.5a}$$

$$\vec{E}_{LC} = (E_0\hat{x} + iE_0\hat{y})e^{-ikz} \tag{5.2.5b}$$

Convert (5.2.5a) and (5.2.5b) into time domain

$$\vec{E}_{RC}(z,t) = E_0[\cos(\omega t - kz)\,\hat{x} + \sin(\omega t - kz)\,\hat{y}] \tag{5.2.6a}$$

$$\vec{E}_{LC}(z,t) = E_0[\cos(\omega t - kz)\,\hat{x} - \sin(\omega t - kz)\,\hat{y}] \tag{5.2.6b}$$

By setting $z = 0$ and plotting the time domain fields we can show that (5.2.6a) is RHCP and (5.2.6b) is LHCP. Note that the convention used in the optics community follows the opposite definition.

C. Elliptical Polarization

Elliptical polarization is a general case that arises when conditions (5.2.3) and (5.2.4) are not satisfied. However, it is important to note that a wave that is neither linearly nor circularly polarized is not necessarily elliptically polarized. Most electromagnetic waves encountered in nature, such as daylight, are unpolarized.

□ **<u>Example 5-3</u>** <u>(Cheng 8-3)</u>

Prove that a linearly polarized plane wave can be decomposed into equal-amplitude right-hand and left-hand circularly polarized waves.

Solution

$$\vec{E} = E_0 e^{-ikz} = \frac{E_0}{2}(\vec{x} - i\vec{y})e^{-ikz} + \frac{E_0}{2}(\vec{x} - i\vec{y})e^{-ikz} = \vec{E}_{RC} + \vec{E}_{LC}$$

5.3 Group Velocity

In (5.1.13) we have defined the phase velocity. Group velocity refers to the speed at which a wave packet or a pulse of waves propagates through a medium. Unlike phase velocity, which describes the speed of an individual wave within the packet, the group velocity represents the speed at which energy or information travels.

Consider a simple wave packet composed of two traveling waves with equal amplitude but slightly different angular frequencies, $\omega_0 + \Delta\omega$ and $\omega_0 - \Delta\omega$, where $\omega_0 \gg \omega$. Since the phase constant depends on frequency,

the corresponding phase constants are $k_0 + \Delta k$ and $k_0 - \Delta k$. The total wave is then:

$$\begin{aligned}\vec{E}(z,t) &= E_0 \cos[(\omega_0 + \Delta\omega)t - (k_0 + \Delta k)z] \\ &\quad + E_0 \cos[(\omega_0 - \Delta\omega)t - (k_0 - \Delta k)z] \\ &= 2E_0 \cos(\Delta\omega t - \Delta k z)\cos(\omega_0 t - k_0 z)\end{aligned} \tag{5.2.1}$$

Since $\omega_0 \gg \omega$, this expression describes a rapidly oscillating wave at frequency $\omega_0$with a slowly varying amplitude modulated by $\Delta\omega$.

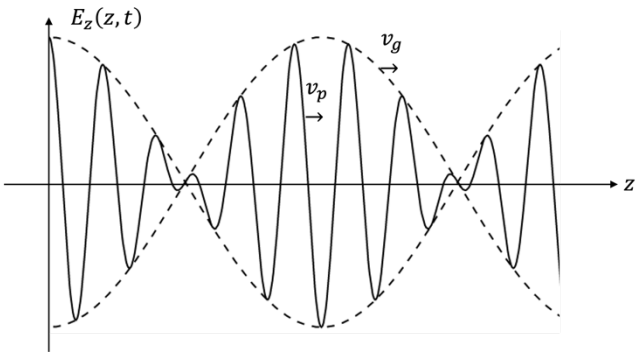


Fig. 5.1
Illustration of phase and group velocity

The wave inside the envelope travels with the phase velocity

$$v_p = \frac{\omega_0}{k_0} \tag{5.2.2}$$

For the velocity of envelope, which is the group velocity v_g, we set the argument in the first cosine in (5.2.1) to be constant and in the limit of $\Delta\omega, \Delta k \to 0$, we get

$$v_g = \frac{\Delta\omega}{\Delta k} \to \frac{d\omega}{dk} \tag{5.2.3}$$

A figurative illustration of phase velocity and group velocity is as the following ω-k graph.

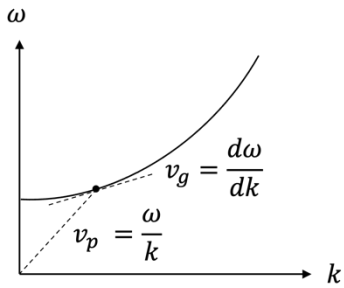


Fig. 5.2
ω-k graph.

Next, we derive the relationship between v_p and v_g

$$v_g = \frac{1}{\frac{dk}{d\omega}} = \frac{1}{\frac{d\left(\frac{\omega}{v_p}\right)}{d\omega}} = \frac{v_p}{1 - \frac{\omega}{v_p}\frac{dv_p}{d\omega}} \tag{5.2.4}$$

Three possible cases are characterized:

i. $\frac{dv_p}{d\omega} = 0$ and $v_g = v_p$: no dispersion
ii. $\frac{dv_p}{d\omega} < 0$ and $v_g < v_p$: normal dispersion
iii. $\frac{dv_p}{d\omega} > 0$ and $v_g > v_p$: anomalous dispersion

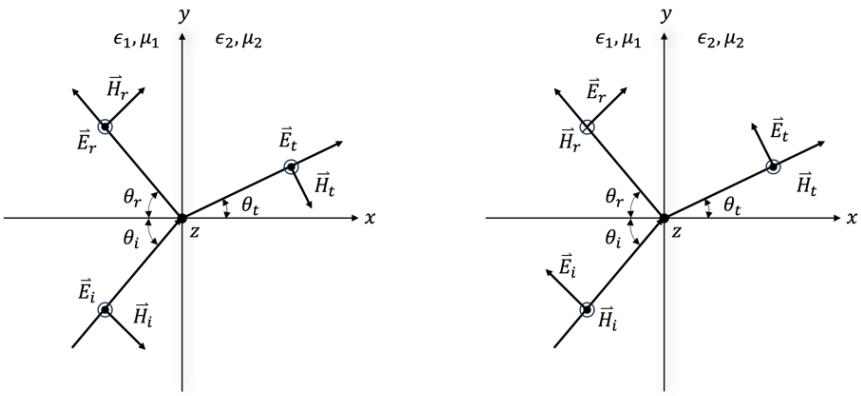
5.4 Transmission and Reflection

In this section, we examine a plane wave incident on a dielectric boundary at an arbitrary angle. Two polarization cases are considered: perpendicular and parallel polarization. Since any arbitrarily polarized plane wave can be decomposed into these two components, analyzing them separately provides a complete description of the wave interaction with plane dielectric boundaries. We are analyzing the relationship between the incident wave and the reflected/transmitted wave. Let us set the amplitude of the incident electric field to be E_i, and define the reflection/transmission coefficient as

$$\Gamma = \frac{E_r}{Ei} \tag{5.4.1}$$

$$\mathrm{T} = \frac{E_t}{Ei} \tag{5.4.2}$$

where E_r, E_t are the amplitudes of the reflected/transmitted electric fields.



(a) (b)

Fig. 5.3 (a) perpendicular polarization, (b) parallel polarizarion.

5.4.1 Oblique Incidence - Perpendicular Polarization

Perpendicular polarization (also called s-polarization) refers to the case where the electric field is perpendicular to the plane of incidence. In this case, the magnetic field lies in the plane of incidence. From (5.1.9) and (5.1.11), the incident electric field and magnetic field can be expressed as

$$\vec{E}_i = E_i e^{-i\vec{k}_i \cdot \vec{r}} \hat{z} = E_i e^{-ik_1(x\cos\theta_i + y\sin\theta_i)} \hat{z} \tag{5.4.3a}$$

$$\vec{H}_i = \frac{1}{\eta_1} \hat{k}_i \times \vec{E}_i$$

$$= \frac{1}{\eta_1} (\cos\theta_i \, \hat{x} + \sin\theta_i \, \hat{y}) \times E_i e^{-ik_1(x\cos\theta_i + y\sin\theta_i)} \hat{z} \tag{5.4.3b}$$

$$= \frac{E_i}{\eta_1} e^{-ik_1(x\cos\theta_i + y\sin\theta_i)}(\sin\theta_i\,\hat{x} - \cos\theta_i\,\hat{y})$$

Accordingly, the reflected fields can be expressed as

$$\vec{E}_r = \Gamma E_i e^{-ik_1(-x\cos\theta_r + y\sin\theta_r)}\hat{z} \tag{5.4.4a}$$

$$\vec{H}_r = \frac{\Gamma E_i}{\eta_1} e^{-ik_1(-x\cos\theta_i + y\sin\theta_i)}(\sin\theta_i\,\hat{x} + \cos\theta_i\,\hat{y}) \tag{5.4.4b}$$

Also, the transmitted fields can be expressed as

$$\vec{E}_t = \mathrm{T} E_i e^{-ik_2(x\cos\theta_t + y\sin\theta_t)}\hat{z} \tag{5.4.5a}$$

$$\vec{H}_t = \frac{\mathrm{T} E_i}{\eta_2} e^{-ik_2(x\cos\theta_t + y\sin\theta_t)}(\sin\theta_t\,\hat{x} - \cos\theta_t\,\hat{y}) \tag{5.4.5b}$$

Applying the matching conditions:

$$\left(\vec{E}_i + \vec{E}_r\right)\big|_{\tan,\ x=0} = \left(\vec{E}_t\right)\big|_{\tan,\ x=0} \tag{5.4.6a}$$

$$\left(\vec{H}_i + \vec{H}_r\right)\big|_{\tan,\ x=0} = \left(\vec{H}_t\right)\big|_{\tan,\ x=0} \tag{5.4.6b}$$

we have

$$e^{-ik_1 y\sin\theta_i} + \Gamma e^{-ik_1 y\sin\theta_r} = \mathrm{T} e^{-ik_2 y\sin\theta_t} \tag{5.4.7a}$$

$$\frac{1}{\eta_1}\left(-\cos\theta_i\, e^{-ik_1 y\sin\theta_i} + \Gamma\cos\theta_r\, e^{-ik_1 y\sin\theta_r}\right) = \frac{-1}{\eta_2}\mathrm{T}\cos\theta_t\, e^{-ik_2 y\sin\theta_t} \tag{5.4.7b}$$

Since (5.4.7) holds for all values of y, the term on the exponentials should be the same for all y. Thus, we have the phase matching condition

$$k_1\sin\theta_i = k_1\sin\theta_r = k_2\sin\theta_t \tag{5.4.8}$$

or, when non-magnetic materials are considered

$$n_1 \sin\theta_i = n_2 \sin\theta_t \tag{5.4.9}$$

which is also known as the Snell's law. n_1 and n_2 are the refractive index of the materials, having the definition of

$$n = \sqrt{\epsilon_r} = \sqrt{\epsilon/\epsilon_0} \tag{5.4.10}$$

For reflection angle, we have

$$\theta_i = \theta_r \tag{5.4.11}$$

Using (5.4.8), we further reduce (5.4.7) to

$$1 + \Gamma = \mathrm{T} \tag{5.4.12a}$$

$$\frac{\cos\theta_i}{\eta_1}(-1 + \Gamma) = \frac{-\cos\theta_t}{\eta_2}\mathrm{T} \tag{5.4.12b}$$

Solving (5.4.12), we have

$$\Gamma = \frac{\eta_2/\cos\theta_t - \eta_1/\cos\theta_i}{\eta_2/\cos\theta_t + \eta_1/\cos\theta_i} \tag{5.4.13a}$$

$$\mathrm{T} = \frac{2\eta_2/\cos\theta_t}{\eta_2/\cos\theta_t + \eta_1/\cos\theta_i} \tag{5.4.13b}$$

$\eta_1/\cos\theta_i$ and $\eta_2/\cos\theta_t$ represent the ratio of the tangential components of the electric field to the magnetic field.

5.4.2 Oblique Incidence - Parallel Polarization

Parallel polarization (also called p-polarization) refers to the case where the electric field is parallel to the plane of incidence. Similar to the analysis of perpendicular polarization, we can express the incident fields as

$$\vec{E}_i = E_i e^{-ik_1(x\cos\theta_i + y\sin\theta_i)}(-\sin\theta_i\,\hat{x} + \cos\theta_i\,\hat{y}) \tag{5.4.14a}$$

$$\vec{H}_i = \frac{E_i}{\eta_1} e^{-ik_1(x\cos\theta_i + y\sin\theta_i)}\hat{z} \tag{5.4.14b}$$

Accordingly, the reflected fields can be expressed as

$$\vec{E}_r = \Gamma E_i e^{-ik_1(-x\cos\theta_r + y\sin\theta_r)}(\sin\theta_i\,\hat{x} + \cos\theta_i\,\hat{y}) \tag{5.4.15a}$$

$$\vec{H}_r = -\frac{\Gamma E_i}{\eta_1} e^{-ik_1(-x\cos\theta_i + y\sin\theta_i)}\hat{z} \tag{5.4.15b}$$

Also, the transmitted fields can be expressed as

$$\vec{E}_t = \mathrm{T} E_i e^{-ik_2(x\cos\theta_t + y\sin\theta_t)}(-\sin\theta_i\,\hat{x} + \cos\theta_i\,\hat{y}) \tag{5.4.16a}$$

$$\vec{H}_t = \frac{\mathrm{T} E_i}{\eta_2} e^{-ik_2(x\cos\theta_t + y\sin\theta_t)}\hat{z} \tag{5.4.16b}$$

Applying the matching conditions (5.4.6) and Snell's law (5.4.8), we get the following expression for the reflection and transmission coefficients

$$\Gamma = \frac{\eta_2\cos\theta_t - \eta_1\cos\theta_i}{\eta_2\cos\theta_t + \eta_1\cos\theta_i} \tag{5.4.17a}$$

$$\mathrm{T} = \frac{2\eta_2\cos\theta_t}{\eta_2\cos\theta_t + \eta_1\cos\theta_i}\left(\frac{\cos\theta_i}{\cos\theta_t}\right) \tag{5.4.17b}$$

and the relationship

$$1 + \Gamma = \mathrm{T}\left(\frac{\cos\theta_t}{\cos\theta_i}\right) \tag{5.4.18}$$

The factor $\left(\frac{\cos\theta_i}{\cos\theta_t}\right)$ in (5.4.17b) comes from the definition T being the ratio of the total electric field $\mathrm{T} = \frac{E_t}{E_i} = \frac{E_{yt}/\cos\theta_t}{E_{yi}/\cos\theta_i} = \frac{E_{yt}}{E_{yi}}\left(\frac{\cos\theta_i}{\cos\theta_t}\right)$.

5.4.3 Special Cases

This chapter focuses on the oblique incidence of plane waves between dielectrics. For normal incidence, setting the incident angle to zero ensures

that Sections 5.4.1 and 5.4.2 yield the same results. For example, for normal incidence, we have

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \tag{5.4.19a}$$

$$\mathrm{T} = \frac{2\eta_2}{\eta_2 + \eta_1} \tag{5.4.20b}$$

If medium 2 is PEC, we can set $\epsilon_2 = \epsilon_c \to i\infty$ and $\eta_2 \to 0$. This implies that there is no field transmitted into the PEC with $\mathrm{T} = 0$.

□ **<u>Example 5-4</u>** <u>(Cheng 8-9)</u>

A y-polarized uniform plane wave $(\vec{E}_i, \vec{H}_i)$ with a frequency of 100 MHz propagates in air along the +x-direction and is normally incident on a perfectly conducting plane at $x = 0$. Given that the amplitude of E_y is 6×10^{-3} V/m,

(a) Express $(\vec{E}_i, \vec{H}_i)$ for the incident wave.
(b) Determine the reflected fields $(\vec{E}_r, \vec{H}_r)$.
(c) Determine the total fields $(\vec{E}, \vec{H})$ in air.
(d) Find the locations to the conducting plane where $\vec{E}$ is zero.

Solution

$$\omega = 2\pi f = 2\pi \times 10^8$$
$$k_1 = \frac{\omega}{c} = \frac{2\pi}{3}$$
$$\eta_1 = 120\pi$$

(a)

$$\vec{E}_i = E_i e^{-ik_1 x}\,\hat{y} = 6 \times 10^{-3} e^{-i\frac{2\pi}{3}x}\,\hat{y}$$
$$\vec{H}_i = \frac{1}{\eta_1}\hat{x} \times \vec{E}_i = \frac{10^{-4}}{2\pi} e^{-i\frac{2\pi}{3}x}\,\hat{z}$$

(b)

For normal incident on a PEC, $\Gamma = \frac{0\cos 0 - \eta_1 \cos 0}{0\cos 0 + \eta_1 \cos 0} = -1$

$$\vec{E}_r = -E_i e^{+ik_1 x}\,\hat{y} = -6\times 10^{-3} e^{+i\frac{2\pi}{3}x}\,\hat{y}$$

$$\vec{H}_r = \frac{1}{\eta_1}(-\hat{x})\times\vec{E}_r = \frac{10^{-4}}{2\pi} e^{+i\frac{2\pi}{3}x}\,\hat{z}$$

(c)

$$\vec{E} = \vec{E}_i + \vec{E}_r = -i12\times 10^{-3}\sin\frac{2\pi x}{3}\,\hat{y}$$

$$\vec{H} = \vec{H}_i + \vec{H}_r = \frac{10^{-4}}{\pi}\cos\frac{2\pi x}{3}\,\hat{z}$$

which is clearly a standing wave.

(d)

Null occurs at $\sin\frac{2\pi x}{3} = 0$, i.e., $x = -\frac{3}{2}n, \quad n = 0, 1, 2, 3 \ldots$

5.4.4 Critical Angle and Brewster's Angle

Two notable cases occur when total reflection and zero reflection take place. The incident angle at which total reflection occurs is called the critical angle, while the angle at which zero reflection occurs is known as Brewster's angle.

A. Critical angle

Total reflection occurs when $\theta_t = 90°$, which means from (5.4.8)

$$\sqrt{\epsilon_1\mu_1}\sin\theta_c = \sqrt{\epsilon_2\mu_2} \tag{5.4.20}$$

or,

$$\theta_c = \sin^{-1}\left(\sqrt{\epsilon_2\mu_2/\epsilon_1\mu_1}\right) \tag{5.4.21}$$

where θ_c is called the critical angle. We encounter total reflection when $\theta_i \geq \theta_c$.

For (5.4.21) to have a positive real-valued solution, $0 < \sqrt{\epsilon_2\mu_2/\epsilon_1\mu_1} < 1$. For non-magnetic materials $\mu_1 = \mu_2$, we have the following relation

$$n_2 < n_1 \tag{5.4.22}$$

where $n = \sqrt{\epsilon}$ is the refractive index. It is noted that (5.4.8) holds for both perpendicular and parallel polarizations, thus critical angle can exist for both perpendicular and parallel polarizations.

B. Brewster's angle

Zero reflection occurs when (5.4.11a) or (5.4.17a) equal to zero. For the perpendicular polarization $\eta_2/\cos\theta_t = \eta_1/\cos\theta_i$, we will have

$$(1 - \sin^2\theta_i) = \frac{\mu_1\epsilon_2}{\epsilon_1\mu_2}(1 - \sin^2\theta_t) \tag{5.4.23}$$

From Snell's law $\sin^2\theta_t = \frac{\mu_1\epsilon_1}{\mu_2\epsilon_2}\sin^2\theta_i$, we get

$$\sin\theta_i = \sqrt{\frac{\frac{\epsilon_2}{\epsilon_1} - \frac{\mu_2}{\mu_1}}{\frac{\mu_1}{\mu_2} - \frac{\mu_2}{\mu_1}}} \tag{5.4.24}$$

For θ_i in (5.4.24) to have real solution, the left-hand side can not exceed unity

$$\frac{\epsilon_2}{\epsilon_1} \leq \frac{\mu_1}{\mu_2} \tag{5.4.25}$$

However, for most materials $\mu_1 = \mu_2$, the left-hand side of (5.4.24) is infinity, which implies that there exists no Brewster's angle for these materials.

For the parallel polarization $\eta_2\cos\theta_t = \eta_1\cos\theta_i$, we will have

$$(1 - \sin^2\theta_i) = \frac{\mu_2\epsilon_1}{\epsilon_2\mu_1}(1 - \sin^2\theta_t) \tag{5.4.26}$$

From Snell's law $\sin^2\theta_t = \frac{\mu_1\epsilon_1}{\mu_2\epsilon_2}\sin^2\theta_i$, we get

$$\sin\theta_i = \sqrt{\frac{\frac{\epsilon_2}{\epsilon_1} - \frac{\mu_2}{\mu_1}}{\frac{\epsilon_2}{\epsilon_1} - \frac{\epsilon_1}{\epsilon_2}}} \tag{5.4.27}$$

For θ_i in (5.4.27) to have real solution, the left-hand side sould not exceed unity

$$\frac{\epsilon_1}{\epsilon_2} \leq \frac{\mu_2}{\mu_1} \tag{5.4.28}$$

For most materials $\mu_1 = \mu_2$, (5.4.27) reduces to

$$\begin{aligned}\theta_i = \theta_b &= \sin^{-1}\left(\sqrt{\frac{\epsilon_2}{\epsilon_1 + \epsilon_2}}\right) \\ &= \cos^{-1}\left(\sqrt{\frac{\epsilon_1}{\epsilon_1 + \epsilon_2}}\right) = \tan^{-1}\left(\sqrt{\frac{\epsilon_2}{\epsilon_1}}\right)\end{aligned} \tag{5.4.29}$$

□ **<u>Example 5-5</u>** <u>(Cheng 8-13)</u>

The permittivity of water at optical frequencies is $1.75\epsilon_0$. An isotropic light source located at a depth d underwater produces an illuminated circular area with a radius of 5 m. Determine d.

Solution

$$\theta_c = \sin^{-1}\left(\sqrt{\frac{\epsilon_0}{1.75\epsilon_0}}\right) = \sin^{-1}\left(\frac{1}{1.32}\right) = 49.2^\circ$$

$$d = \frac{5}{\tan(49.2^\circ)} = 4.32 \text{ (m)}$$